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 SHARDA UNIVERSITY <i>Beyond Boundaries</i>		School of Computing Science & Engineering Department of Computer Science & Engineering				
Assignment-1, Session:2026-27 [Programme: B Tech] [Semester: 7th]						
Course Title: Mobile Computing Course Code: CSE062				Max Marks: 20		
Instructions:		All questions are compulsory Assume missing data suitably, if any All questions should be of higher learning.			BTL Level (K)	
Unit 1		CO1: Synthesize the basic concepts and principles in mobile computing.				
Unit 2		CO2: Analyze the concept of wireless & telecommunication networks.				
Q.No.	Section A (4×3=12 Marks)			Unit	COs	K
1	Explain the need for wireless transmission in mobile computing. Discuss the factors that affect the selection of radio transmission frequencies for a mobile communication system.			Unit 1	CO1	L2
2	A mobile system must support users moving through different coverage areas. Compare FDMA, TDMA and CDMA in terms of channel allocation, interference and suitability for cellular communication. Justify which technique provides better multiple-access capability.			Unit 1	CO1	L4
3	Explain the basic architecture of a GSM network and describe the role of the major components involved in providing mobile services.			Unit 2	CO2	L2
4	A mobile user moves from one cell to another while an active call is in progress. Explain the handover process and analyze why handover is essential for maintaining service continuity.			Unit 2	CO2	L4
Section B (2×4=8 Marks)						
5	A wireless network is designed for many mobile users. Evaluate the suitability of SDMA, FDMA, TDMA and CDMA for sharing the wireless medium. Recommend an appropriate multiple-access scheme for a cellular network and justify your choice.			Unit 1	CO1	L5
6	Consider a mobile data service using GPRS. Analyze the GPRS architecture and explain how its network nodes support packet-based data communication. Compare packet-based GPRS communication with traditional circuit-switched mobile communication.			Unit 2	CO2	L4

Unit	Questions Numbers	Total Marks	CO1	CO2
1	1,2,5	10	10	-
2	3,4,6	10	-	10